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Project Title: BRAC Bank PLC Database Audit, Normalization and Business Intelligence Reporting

1. Introduction & Business Context

BRAC Bank PLC is one of Bangladesh's leading private commercial banks, offering retail banking, SME banking, corporate banking, and treasury services. With a large customer base and multiple branches nationwide, the bank relies heavily on its transactional databases for daily operations, reporting, and regulatory compliance. However, the existing database contained structural anomalies, data quality issues, and normalization violations, which impacted performance, reporting accuracy, and data integrity. This project conducts a full database audit, applies corrections, normalizes tables to 2NF, and prepares the database for Business intelligence reporting.

2. Initial Database Issues & Applied Fixes

2.1 Data Cleaning & Quality Fixes

Issue	Issue Details	Issue Type	Applied Fix	Solution Details
Inconsistent City Formatting	City names contained extra spaces and inconsistent casing	Data Quality	Case Standardization	Used LTRIM, RTRIM, UPPER to standardize City names
Missing Customer Emails	Customer email fields were NULL	Missing Data	Default Value Imputation	Updated NULL emails to standardize placeholder
Invalid Email Domains	Emails used fake or non-standard domains	Data Integrity	Domain Standardization	Replaced invalid domains with realistic providers
Missing Account Numbers	Account numbers missing in ATM	Missing Data	Default Value Assignment	Assigned default account number pattern

	and bill payment tables			
Missing Transaction Amounts	Amount fields contained NULL values	Missing Data	Zero Imputation	Updated NULL amounts to 0 for consistency
Missing Status Values	Status fields were NULL across multiple tables	Missing Data	Default Status Assignment	Set missing statuses to 'Pending' or 'Active'
Extra Spaces in Text Columns	Name, phone, email, address contained leading/trailing spaces	Data Quality	String Trimming	Applied LTRIM and RTRIM
Inconsistent Email Casing	Emails stored in mixed case	Data Quality	Lowercase Conversion	Used LOWER() function
Invalid Date Strings	Dates stored as VARCHAR with inconsistent formats	Format Error	Date Conversion	Added new DATE/DATETIME columns and populated using direct assignment
Non-Numeric Amount Values	Amount column contained non-numeric strings	Data Quality	Zero Replacement	Set non-numeric values to 0
Invalid Phone Numbers	Phone numbers without country code or extra spaces	Format Error	Standardization	Used LTRIM and RTRIM
Duplicate Account Numbers	Same account_number appeared multiple times	Data Redundancy	Duplicate Removal	Kept minimum account_id per account_number

2.2 Referential Integrity & Orphan Records

Issue	Issue Details	Issue Type	Applied Fix	Solution Details
Transactions Without Accounts	Transaction records referenced missing account numbers	Integrity Violation	Orphan Validation & FK Enforcement	Validated references then added FK constraints
Cheque Transactions Without Accounts	Cheque records had wrong account number prefix	Integrity Violation	Prefix Correction	Corrected 'EB' prefix to 'BBL'
Duplicate Fraud Alerts	Multiple alerts for same transaction ID	Data Redundancy	Duplicate Detection	Identified 8 repetitive records
Duplicate Customer Records	Same email or phone assigned to multiple customers	Data Integrity	Unique Constraint Addition	Added unique constraints on phone and NID
Orphan Branch Performance	Performance records with invalid branch references	Integrity Violation	Foreign Key Enforcement	Added FK to Branches_backup
Orphan Loan Payments	Payment records referencing non-existent loans	Integrity Violation	Foreign Key Enforcement	Added FK to Loans_backup

2.3 Data Type Standardization

Issue	Issue Details	Issue Type	Applied Fix	Solution Details
Amount as String	Amount columns stored as VARCHAR	Structural Issue	Decimal Conversion	Converted to DECIMAL(18,2)
Date as String	Date columns stored as text	Format Error	Date Type Conversion	Converted to DATE or DATETIME
Primary Key Mixed Types	PK columns had inconsistent data types	Structural Issue	INT Enforcement	Converted all PKs to INT NOT NULL
Foreign Key Type Mismatch	FK columns did not match referenced PK types	Structural Issue	Type Alignment	Altered FK columns to match PK types
Boolean Stored as String	Yes/No fields stored as text	Data Quality	Bit Conversion	Converted to BIT data type
Interest Rate Precision	Interest rates had inconsistent decimal places	Format Error	Decimal Standardization	Set to DECIMAL(5,2)
Redundant Currency Column	Currency stored unnecessarily	Normalization Issue	Column Removal	Dropped redundant currency column

2.4 Date Standardization

Issue	Issue Details	Issue Type	Applied Fix	Solution Details
Date Stored as String	Transaction, payment, effective, maturity, and logged dates stored as VARCHAR across multiple tables	Format Error	Date type Conversion	Added new DATE/DATETIME columns and populated using direct assignment
Invalid Date Values	Some dates failed conversion during cleaning	Data Quality	Audit & Validation	Used audit queries to identify and flag failures
Redundant Date Columns	Month stored separately after extraction from month_year	Normalization Issue	Column Removal	Dropped redundant month column after linking to Month_Name lookup

3. Normalization (Up to 2NF)

3.1 Lookup & Benchmark Tables (Domain Integrity)

Table	Issue	Applied Fix
Accounts_backup	Account type and status as raw text	Created Account_Type and Account_Status lookup tables
ATM_Transactions_backup	Transaction type, location, status as free text	Created ATM_Transaction_Type, ATM_Location, ATM_Status lookup tables

BillPayments_backup	Bill type and status as plain text	Created Bill_Type and Bill_Status lookup tables
Branches_backup	City stored as string	Created City lookup table
BranchPerformance_backup	Month stored without reference	Created Month_Name lookup table
ChequeTransactions_backup	Status as free text	Created Cheque_Status lookup table
CreditCards_backup	Card type and status as raw text	Created Card_Type and Card_Status lookup tables
CreditCardTransactions_backup	Transaction status and category as text	Created CC_Status and Category lookup tables
Customers	Customer status as raw string	Created Customer_Status lookup table
Employees	Department, designation, status as free text	Created Departments, Designations, Employee_Status lookup tables
InterestRates	Account type as text without FK	Referenced existing Account_Type table
KYCDocuments	Document type as raw string	Created KYC_Doc_Type lookup table
LoanPayments_backup	Payment method and status as text	Created Loan_Payment_Method and LoanPayment_Status lookup tables
Loans_backup	Loan type and status as text	Created Loan_Type and Loan_Status lookup tables
Promotions_backup	Applicable account types as text	Referenced Account_Type table
SupportTickets_backup	Issue type as free text	Created Support_Issue_Type lookup table
Transactions_backup	Transaction type, status, channel as free text	Used Transaction_Type, created Transaction_Status and Transaction_Channel lookup tables

3.2 Multi-Valued Attribute Handling

Issue	Applied Fix
Customer name stored as single column	Split into first_name and last_name
Employee name stored as single column	Split into first_name and last_name
Month-year stored as combined string	Split into year and month_id, linked to Month_Name lookup table

4. Data Type Standardization & Key Readiness

4.1 Numeric & Monetary Fixes

Issue	Applied Fix
Amount columns unsuitable for math	Converted to DECIMAL(18,2)
Interest rates with inconsistent precision	Standardized to DECIMAL(5,2)
Integer columns with NULL values	Set NOT NULL with default values
Boolean values stored as text	Converted to BIT data type

4.2 Primary Key Standardization:

All identifier columns were standardized to INT NOT NULL and assigned explicit Primary Key constraints:

Tables covered: Accounts_backup, ATM_Transactions_backup, BillPayments_backup, Branches_backup, BranchPerformance_backup, ChequeTransactions_backup, CreditCards_backup, CreditCardTransactions_backup, CreditScores, CustomerTypes, Customers, Employees, FixedDeposits, FraudAlerts, InterestRates, KYC_backup, LoanPayments_backup, Loans_backup, Promotions_backup, SupportTickets_backup, TransactionsTypes_backup, Transactions_backup

Business Importance:

- Guarantees row uniqueness across all entities
- Required foundation for foreign key constraints
- Prevents duplicate and ambiguous records
- Enables reliable joins for BI reporting

4.3 Unique Constraints Added

Table	Unique Constraint	Purpose
Accounts_backup	Account_Number	Prevents duplicate account numbers
Branches_backup	Branch_Name	Ensures branch name uniqueness
Customers	Phone and NID	Prevents duplicate customer records

5. Foreign Key Assignment (Relationship Enforcement)

Relationship	Purpose
Accounts_backup → Customers	Ensures every account belongs to a valid customer
Accounts_backup → Branches_backup	Ensures account is linked to existing branch
Customers → CustomerTypes	Enforces valid customer type reference
Customers → Customer_Status	Validates customer status against lookup table
ATM_Transactions_backup Accounts_backup →	Ensures ATM transactions reference valid accounts
BillPayments_bakup → Accounts_backup	Validates bill payment account references
ChequeTransactions_backup Accounts_backup →	Ensures cheque transactions have valid accounts
Transactions_backup → Transaction_Type	Validates transaction type against lookup
Transactions_backup → Transaction_Status	Enforces valid status reference
Transactions_backup Transaction_Channel →	Validates channel against lookup
CreditCards_backup → Customers	Ensures credit cards belong to valid customers
CreditCards_backup → Accounts_backup	Links credit cards to valid accounts
CreditCards_backup → Card_Type	Validates card type classification
CreditCards_backup → Card_Status	Enforces valid status reference
CreditCardTransactions_backup Category →	Validates transaction category

CreditCardTransactions_backup CC_Status	→	Ensures valid transaction status
Loans_backup	→ Customers	Validates loan customer ownership
Loans_backup	→ Accounts_backup	Links loans to valid accounts
Loans_backup	→ Loan_Type	Enforces valid loan type classification
Loans_backup	→ Loan_Status	Validates loan status
LoanPayments_backup	→ Loans_backup	Ensures payments reference valid loans
LoanPayments_backup Loan_Payment_Method	→	Validates payment method
LoanPayments_backup LoanPayment_Status	→	Enforces valid payment status
FixedDeposits	→ Customers	Validates FD customer ownership
FixedDeposits	→ Accounts_backup	Links FDs to valid accounts
InterestRates	→ Branches_backup	Links interest rates to valid branches
InterestRates	→ Account_Type	Validates account type for interest rate
SupportTickets_backup	→ Customers	Validates ticket customer ownership
SupportTickets_backup Accounts_backup	→	Links tickets to valid accounts
SupportTickets_backup Support_Issue_Type	→	Enforces valid issue type classification
KYC_backup	→ Customers	Ensures KYC documents belong to valid customers
KYC_backup	→ KYC_Doc_Type	Validates document type classification
Employees	→ Branches_backup	Ensures employees belong to valid branches

Employees → Departments	Validates department assignment
Employees → Designations	Enforces valid designation
Employees → Employee_Status	Validates employee status
BranchPerformance_backup → Branches_backup	Ensures performance records link to valid branches
BranchPerformance_backup → Month_Name	Validates month reference
Promotions_backup → Account_Type	Ensures promotions apply to valid account types
FraudAlerts → Accounts_backup	Links fraud alerts to valid accounts

6. Analytical Reports & Insights:

Report 1: Active vs Closed Accounts Overview (Operational Performance)

Report Name: Active vs Closed Accounts Overview

Report Summary: This query shows how many accounts BRAC Bank currently has in each status (Active, Closed, Dormant) and the total balance held in each category, giving a snapshot of customer engagement and deposit retention.

Query:

```
select
s.status_name as account_status,
count(*) as total_accounts,
sum(a.balance) as total_balance
from Accounts_backup a
join Account_Status s on a.status = s.status_id
group by s.status_name
order by total_accounts desc;
```

Query Output:

	account_status	total_accounts	total_balance
1	Active	2751	6953423916.73
2	Dormant	2633	6733696536.68
3	Closed	2616	6522222724.91

Report Explanation:

- **What it shows:** Number of accounts and total money in Active, Closed, and Dormant status.
- **Why it matters:** Active accounts generate revenue through fees, transactions, and loans. Closed accounts represent lost customers. Dormant accounts are inactive but still cost the bank maintenance expenses.
- **How it helps management or operations:** Operations team can identify if dormant accounts are growing and take action. Management can track customer attrition over time.

Link technical findings to business decisions:

- `sum(a.balance)` for Closed accounts shows how much money left the bank. If this is high, customers are taking large balances elsewhere.
- `count(*)` for Dormant accounts helps calculate the operational cost of maintaining inactive accounts.

Business Value & Recommendation:

The output shows the health of the customer base. If Dormant accounts exceed 15% of total, recommend: launch reactivation campaign (waive dormancy fees, send SMS alerts). If Closed accounts are growing monthly, recommend: conduct exit surveys to understand why customers leave and fix root causes.

Report 2: Total Deposits by Branch (Financial Trend)

Report Name: Total Deposits by Branch

Report Summary: This query shows which branches hold the most customer deposits and their city distribution, helping management identify high-performing and underperforming locations.

Query:

```
select
b.branch_name,
c.city_name as city,
sum(a.balance) as total_deposits
From Accounts_backup a
Join Branches_backup b on a.branch_id =
b.branch_id
Join City c on b.city = c.city_id
Group by b.branch_name, c.city_name
Order by total_deposits desc;
```

Query Output:

	branch_name	city	total_deposits
1	Chittagong Main Branch	Chittagong	3430037712.85
2	Dhaka Main Branch	Dhaka	3419514144.78
3	Rajshahi Branch	Rajshahi	3383293110.88
4	Gulshan Branch	Dhaka	3372492260.21

5	Khulna Branch	Khulna	3324840997.48
6	Sylhet Branch	Sylhet	3279164952.12

Report Explanation:

- **What it shows:** Each branch name, its city, and total customer deposits held at that branch.
- **Why it matters:** Deposits are the raw material for bank lending. Branches with higher deposits can issue more loans and generate more interest income.
- **How it helps management or operations:** Regional managers can compare branch performance. Top branches can be rewarded. Bottom branches can be investigated for low footfall, poor service, or strong local competition.

Link technical findings to business decisions:

- ORDER BY total_deposits DESC ranks branches from best to worst. Management can set deposit targets based on top branch performance.
- City-level grouping shows if certain cities outperform others, indicating regional economic strength or marketing effectiveness.

Business Value & Recommendation:

The output shows deposit concentration. If Dhaka branches hold 70% of total deposits but Chittagong has 30% of branches, recommend: shift marketing budget to Chittagong or open new branches in growing cities. If a specific branch has very low deposits despite good location, recommend: audit branch manager performance and customer service quality.

Report 3: Customer Distribution by Account Type (Customer Behavior)

Report Name: Customer Distribution by Account Type

Report Summary: This query shows which account types customers prefer (Savings, Current, Credit, Fixed Deposit) and the average balance per account type, revealing customer saving and spending behavior.

Query:

```
select
t.type_name as account_type,
count(*) as number_of_accounts,
avg(a.balance) as avg_balance
from Accounts_backup a
join Account_Type t on a.account_type = t.type_id
group by t.type_name
order by number_of_accounts desc;
```

Query Output:

	account_type	number_of_accounts	avg_balance
1	Current	2035	2513535.009950
2	Fixed Deposit	2010	2553577.604860
3	Credit	1987	2535301.262506
4	Savings	1968	2502014.653810

Report Explanation:

- **What it shows:** How many accounts of each type exist and the average balance customers keep in each type.
- **Why it matters:** Savings accounts provide stable, low-cost funds. Current accounts are transactional with lower balances but generate fee income. Fixed Deposits lock money for longer periods. Credit cards generate interest and merchant fees.

- **How it helps management or operations:** Product managers can see which accounts are popular and which need promotion. High average balance in Fixed Deposits indicates customers trust the bank for long-term savings.

Link technical findings to business decisions:

- avg(a.balance) per account type. If the average savings balance is low but account count is high, customers use the bank for small savings only.
- count(*) for Credit accounts. If low, customers may prefer competitor credit cards with better rewards.

Business Value & Recommendation:

The output shows product mix. If Savings dominates (70%+ of accounts) but Fixed Deposits have the highest average balance, recommend: run campaigns converting large Savings balances to higher-interest Fixed Deposits. If Credit accounts are very few, recommend: launch a promotional credit card with zero annual fee for the first year and cashback on utility bills.

Report 4: ATM Transaction Success Rate (Operational Efficiency)

Report Name: ATM Transaction Success Rate

Report Summary: This query shows ATM locations and their completed transaction counts, ordered from lowest to highest, identifying ATMs that may have frequent failures or low usage.

Query:

```
select
l.location_name as atm_location,
count(*) as total_transactions
from ATM_Transactions_backup a
join ATM_Status s on a.status = s.status_id
join ATM_Location l on a.atm_location =
l.location_id
where s.status_name = 'Completed'
group by l.location_name
order by total_transactions asc;
```

Query Output:

	atm_location	total_transactions
1	Khulna Branch	621
2	Rajshahi Branch	644
3	Dhaka Main Branch	658
4	Gulshan Branch	664
5	Chittagong Main Branch	677
6	Sylhet Branch	692

Report Explanation:

- **What it shows:** Each ATM location and the number of completed (successful) transactions, ordered from lowest to highest.
- **Why it matters:** ATMs with very few completed transactions may be frequently failing, out of cash, or located in low-traffic areas. This helps identify maintenance priorities.
- **How it helps management or operations:** Operations teams can focus on ATMs that show low completed transaction counts, indicating potential hardware failure, cash shortage, or connectivity issues.

Link technical findings to business decisions:

- ORDER BY total_transactions ASC puts worst-performing ATMs first.

- Using WHERE filter with s.status_name = 'Completed' follows your notes pattern.
- Low completed count could mean either high failure rate OR low foot traffic. Further investigation needed.

Business Value & Recommendation:

The output shows which ATMs have few successful transactions. If Gulshan ATM has 45 completed transactions while Dhaka Main has 1200, Gulshan ATM likely has frequent failures or is located in a poor area. Recommend: dispatch technician to low-performing ATMs immediately, check cash levels, review error logs, and inspect for hardware issues. If an ATM consistently shows low completed count month over month, consider relocating it to a higher-traffic area or replacing the machine.

Report 5: High-Value Customers (Customer Insights)

Report Name: High-Value Customers

Report Summary: This query identifies the top 10 customers by total balance across all their accounts, helping the bank prioritize relationship management for its most valuable clients.

Query:

```
select TOP 10
c.customer_id,
c.first_name,
c.last_name,
sum(a.balance) as total_balance
from Customers c
join Accounts_backup a on c.customer_id =
a.customer_id
group by c.customer_id, c.first_name, c.last_name
order by total_balance desc;
```

Query Output:

	customer_id	first_name	last_name	total_balance
1	4694	Irya	Edwin	30452187.48

2	2542	Chakradhar	Kata	23035544.65
3	3942	Aahana	Dua	22636076.47
4	1742	Agastya	Gade	21391234.38
5	2697	Manthan	Pai	21216130.5
6	1367	Veer	Khare	20145608.8
7	4510	Pushti	Koshy	19932625.92
8	1976	Charvi	Oak	19729972.17
9	40	Rajata	Parekh	19696238.27
10	4322	Omaja	Prashad	19588403.99

Report Explanation:

- **What it shows:** Customer ID, name, and combined balance across all their accounts, showing only the top 10 highest-value customers.
- **Why it matters:** A small number of high-value customers often contribute a large percentage of total deposits. Losing even one to a competitor significantly impacts bank revenue.

- **How it helps management or operations:** Relationship managers can be assigned to these customers. Marketing can offer exclusive benefits. Customer service can prioritize their support tickets.

Link technical findings to business decisions:

- TOP 10 with ORDER BY total_balance DESC isolates the most important customers. Management can create a VIP tier based on this balance threshold.
- sum(a.balance) across multiple accounts shows true customer worth, not just single account balance.

Business Value & Recommendation:

The output shows the bank's most valuable customers. If the top customer has 5 crore BDT and the top 10 collectively hold 30 crore, recommend: assign dedicated relationship managers to each of these top 10 customers. Offer them premium credit cards, lower loan rates, airport lounge access, and priority customer support hotline. Send personalized birthday and holiday greetings. This reduces the likelihood of them leaving for competitor banks.

Report 6: Loan Default Risk Detection (Risk/Anomaly)

Report Name: Loan Default Risk Detection

Report Summary: This query shows total value of defaulted loans versus active loans still being repaid, helping the bank understand its exposure to credit risk.

Query:

```
select
s.status_name as loan_status,
count(*) as total_loans,
sum(l.amount_sanctioned) as total_amount_at_risk
from Loans_backup l
join Loan_Status s on l.status = s.status_id
where s.status_name IN ('Defaulted', 'Active')
group by s.status_name;
```

Query Output:

	loan_status	total_loans	total_amount_at_risk
1	Active	986	25007939750
2	Defaulted	1021	25636729552

Report Explanation:

- **What it shows:** Number of Active loans vs Defaulted loans, and total money sanctioned in each category.
- **Why it matters:** Defaulted loans directly reduce bank profits. High default amounts may indicate weak credit assessment processes or economic downturns affecting borrowers.
- **How it helps management or operations:** Risk management team can see if default rates are increasing. The collections team knows how much money needs recovery.

Link technical findings to business decisions:

- `sum(l.amount_sanctioned)` for Defaulted shows total loss exposure. If this exceeds 5% of the total loan book, the bank may need to increase loan loss provisions.
- Comparing Report 10 shows if cheque bouncers are also loan defaulters.

Business Value & Recommendation:

The output shows credit risk exposure. If the Default amount is 2 crore and Active amount is 50 crore, default rate is 4%. If industry average is 2%, recommend: tighten loan approval criteria, require higher down payments, or increase credit score requirements. If defaulted loans are concentrated in one loan type (e.g., Personal loans from Report 12), recommend: reduce exposure to that product or increase its interest rate to cover higher risk.

Report 7: Monthly Branch Performance Trend (Growth Indicator)**Report Name: Monthly Branch Performance Trend**

Report Summary: This query shows monthly deposit and revenue trends across all branches, helping management identify seasonal patterns and growth trajectories.

Query:

```
select
m.month_name,
sum(bp.total_deposits) as total_deposits,
sum(bp.revenue) as total_revenue
from BranchPerformance_backup bp
join Month_Name m on bp.month_id =
m.month_id
group by m.month_name, m.month_id
order by m.month_id;
```

Query Output:

	month_name	total_deposits	total_revenue
1	January	393428052.26	16486744.79
2	February	246979128.78	12257879.54
3	March	291804305.39	14450096.36
4	April	317098689.95	18462011.42
5	May	377610703.49	11290211.25
6	June	403097356.89	18472515.05
7	July	433102715.02	11554540.73

8	August	384150250.28	14645901.51
9	September	413968474.14	14095293.01
10	October	243221223.81	15429364.93
11	November	307812668.23	18175631.92
12	December	348517941.10	17671251.50

Report Explanation:

- **What it shows:** Each month's total deposits and revenue summed across all branches.
- **Why it matters:** Identifies which months perform best (e.g., bonus season before Eid, December year-end). Helps predict cash flow and plan marketing campaigns.
- **How it helps management or operations:** Finance team can forecast revenue more accurately. Marketing can schedule campaigns before peak months. Operations can ensure sufficient staffing during high-volume periods.

Link technical findings to business decisions:

- ORDER BY month_id ensures chronological order (January to December), not alphabetical.
- Comparing month-over-month growth shows if the bank is growing or declining.

Business Value & Recommendation:

The output shows seasonal patterns. If December and January show highest deposits (year-end bonuses, new year savings goals), recommend: launch "New Year, New Savings" campaign in December with higher interest rates for Fixed Deposits opened in Jan-March. If revenue drops in July-August, recommend: investigate if this is due to reduced transaction volume or seasonal business slowdown, and plan targeted promotions during these months.

Report 8: Most Common Fraud Alert Reasons (Risk Detection)

Report Name: Most Common Fraud Alert Reasons

Report Summary: This query shows fraud reasons with unresolved alerts, helping security teams identify patterns and close vulnerabilities.

Query:

```
select
reason,
count(*) as total_alerts
from FraudAlerts
where resolved = 0
group by reason
order by total_alerts desc;
```

Query Output:

	reason	total_alerts
1	Suspicious Location	109
2	New Device	99
3	Unusual Amount	97
4	High Frequency	88

Report Explanation:

- **What it shows:** Each fraud reason and how many unresolved alerts exist for that reason.
- **Why it matters:** Unresolved fraud alerts represent active risk. Focusing on unresolved alerts rather than total alerts prioritizes work that still needs completion.
- **How it helps management or operations:** Security team knows exactly which fraud types have backlogs. Management can allocate investigation resources to top unresolved categories.

Link technical findings to business decisions:

- WHERE resolved = 0 filters to pending work only (from your notes: WHERE column = value pattern).
- ORDER BY total_alerts DESC puts highest backlog first.

Business Value & Recommendation:

The output shows fraud investigation backlog by reason. If "Suspicious Transaction" has 45 unresolved alerts while other reasons have fewer than 10, recommend: assign two additional fraud analysts to Suspicious Transaction cases. If "Wrong PIN Attempts" has a high unresolved count, recommend: review ATM security footage for possible skimming devices. Set a weekly target to reduce unresolved alerts by 20%.

Report 9: Card Transaction by Category (Customer Spending Behavior)

Report Name: Card Transaction by Category

Report Summary: This query shows where customers spend most using credit cards, helping the bank design targeted rewards programs.

Query

```
select
cat.category_name,
count(*) as transaction_count,
sum(cc.amount) as total_spent
from CreditCardTransactions_backup cc
join Category cat on cc.category = cat.category_id
group by cat.category_name
order by total_spent desc;
```

Query Output:

	category_name	transaction_count	total_spent
1	Food	838	21655933.97
2	Travel	810	21081566.19
3	Bills	814	20915516.34
4	Entertainment	784	19850850.99
5	Shopping	754	19132152.71

Report Explanation:

- **What it shows:** Each spending category, number of credit card transactions, and total money spent.
- **Why it matters:** Understanding customer spending habits allows the bank to offer relevant rewards, partner with popular merchants, and increase card usage.
- **How it helps management or operations:** Product teams can design cashback categories based on actual spending. Marketing can negotiate merchant partnerships. Customer retention improves when card rewards match customer lifestyle.

Link technical findings to business decisions:

- ORDER BY total_spent DESC shows which categories drive most card spending. If Shopping and Travel dominate, prioritize those for rewards.

- transaction_count vs total_spent. High count but low spent indicates many small transactions (e.g., Food, Bills). High spent but low count indicates big purchases (e.g., Travel, Electronics).

Business Value & Recommendation:

The output shows spending patterns. If Shopping is top category with 40% of total spent, recommend: launch "5% cashback on all Shopping purchases" promotion for 3 months. Partner with Daraz, Pickaboo, or local shopping malls for exclusive discounts for BRAC Bank cardholders. If Travel is second highest, recommend: partner with airlines (Biman, US-Bangla) for 10% off flight bookings using BRAC Bank credit cards.

Report 10: Cheque Bounce Rate (Risk Indicator)

Report Name: Cheque Bounce Rate

Report Summary: This query shows the number of cheques in each status, indicating the reliability of customers issuing cheques.

Query:

```
select
s.status_name,
count(*) as total_cheques
from ChequeTransactions_backup c
join Cheque_Status s on c.status = s.status_id
group by s.status_name
order by total_cheques desc;
```

Query Output:

	status_name	total_cheques
1	Cleared	513

2	Pending	503
3	Bounced	502
4	Cancelled	482

Report Explanation:

- **What it shows:** Number of cheques that cleared successfully, bounced (insufficient funds), pending, or cancelled.
- **Why it matters:** High bounce count means many customers issue cheques without sufficient balance. This wastes bank processing time and creates reconciliation issues.
- **How it helps management or operations:** Risk team can identify if bounced cheques are increasing. Operations can charge higher fees for bounced cheques to discourage behavior.

Link technical findings to business decisions:

- ORDER BY total_cheques DESC shows which status has most cheques. If Bounced is second highest, action needed.
- Percentage can be calculated manually: $\text{Bounced} \div \text{Total} \times 100$.

Business Value & Recommendation:

The output shows cheque clearing health. If Bounced cheques count is 800 out of 10,000 total (8% bounce rate), recommend: implement SMS alert 24 hours before cheque clearance if balance is low. Increase bounced cheque fee from 500 BDT to 1000 BDT after second bounce. If certain customers bounce cheques repeatedly, restrict their cheque book issuance.

Report 11: Promotions Used vs Discount Given (Marketing Efficiency)

Report Name: Promotions Used vs Discount Given

Report Summary: This query shows the top 10 promotion codes by times used, along with discount percentage, helping marketing teams measure campaign effectiveness with limited output.

Query:

```
select top 10
promo_code,
description,
discount_percent,
count(*) as times_used
from Promotions_backup
group by promo_code, description,
discount_percent
order by times_used desc;
```

Query Output:

	promo_code	description	discount_percent	times_used
1	BBL1302	Eos dignissimos natus quod distinctio.	5	1
2	BBL1271	Assumenda quia sequi exercitationem vero id.	10	1
3	BBL1240	Atque explicabo pariatur eos non.	15	1

4	BBL124	Quaerat unde iure ab facilis voluptatem.	10	1
5	BBL1209	Corrupti sequi perferendis aperiam ad.	15	1
6	BBL1178	Reiciendis cumque deleniti quod.	5	1
7	BBL1147	Incidunt accusantium tenetur aperiam ipsa commodi.	10	1
8	BBL1116	Veniam ullam necessitatibus quasi explicabo hic.	10	1
9	BBL1085	Recusandae quod commodi.	5	1
10	BBL1054	Harum autem accusantium nostrum.	5	1

Report Explanation:

- **What it shows:** Top 10 promotion codes by number of times used, with their discount percentage and description.
- **Why it matters:** Large output is unreadable. TOP 10 focuses management attention on most popular promotions only.
- **How it helps management or operations:** Marketing team sees which promotions drive most customer action. Underperforming promotions (not in top 10) can be discontinued.

Link technical findings to business decisions:

- TOP 10 with ORDER BY times_used DESC follows your notes pattern for limiting large results.
- If a promotion has high discount_percent but low times_used, customers may not find it valuable.
- If a promotion has low discount_percent but high times_used, customers respond regardless of discount size.

Business Value & Recommendation:

The output shows which promotions are most popular. If "NEWYEAR10" appears in top 3 with 500+ uses, recommend: run similar promotions in future. If any promotion has very low usage (less than 10 times), recommend: discontinue it. If no promotion is being used much (all under 50 uses), recommend: investigate whether promotion codes are being properly communicated to customers via SMS, email, or branch posters.

Report 12: Interest Income Potential from Active Loans

Report Name: Interest Income Potential from Active Loans

Report Summary: This query shows active loans by type with total principal and average interest rate, allowing management to estimate future interest revenue.

Query:

```
select
lt.type_name as loan_type,
count(*) as active_loans,
sum(l.amount_sanctioned) as total_principal,
avg(l.interest_rate) as avg_interest_rate
from Loans_backup l
join Loan_Type lt on l.loan_type = lt.type_id
join Loan_Status ls on l.status = ls.status_id
where ls.status_name = 'Active'
group by lt.type_name
order by total_principal desc;
```

Query Output:

	loan_type	active_loans	total_principal	avg_interest_rate
1	Education	162	4616590621	11.263024
2	Home	177	4293091384	11.395932
3	Personal	164	4066790779	11.462378
4	Agriculture	153	4063751630	11.492287
5	Business	161	4051638467	11.644472
6	Car	169	3916076869	11.441715

Report Explanation:

- **What it shows:** For each loan type — number of active loans, total principal amount lent, and average interest rate.
- **Why it matters:** Banks earn interest income on active loans. Higher principal and higher interest rates generate more revenue.
- **How it helps management or operations:** Finance team can forecast interest income by multiplying `total_principal` by `avg_interest_rate` manually. Product managers can see which loan types contribute most to bank revenue.

Link technical findings to business decisions:

- ORDER BY total_principal DESC shows which loan types have the largest money exposure.
- Using only sum() and avg() follows your notes exactly.
- Interest income approximation = total_principal × (avg_interest_rate ÷ 100).

Business Value & Recommendation:

The output shows loan portfolio composition. If Home loans have highest total_principal (50 crore) but Personal loans have higher avg_interest_rate (15% vs 9%), calculate: Home interest ≈ 4.5 crore, Personal interest ≈ 1.5 crore. Even with lower rate, Home loans generate more absolute income due to larger principal. Recommend: continue promoting Home loans for stable, long-term income. If Business loans have low avg_interest_rate (7%), consider raising rates to 9-10%.

Report 13: Employee Distribution & Salary Cost by Department

Report Name: Employee Distribution & Salary Cost by Department

Report Summary: This query shows how many active employees work in each department, their total monthly salary cost, and average salary per department, helping HR optimize staffing and budget allocation.

Query:

```
select
d.department_name,
count(*) as total_employees,
sum(e.salary) as total_monthly_salary_cost,
avg(e.salary) as avg_salary_per_employee
from Employees e
join Departments d on e.department_id =
d.department_id
join Employee_Status es on e.status_id =
es.status_id
where es.status_name = 'Active'
group by d.department_name
order by total_monthly_salary_cost desc;
```

Query Output:

	department_name	total_employees	total_monthly_salary_cost	avg_salary_per_employee
1	Front Desk	39	4851255.85	124391.1756
2	Operations	39	4713049.93	120847.4341
3	HR	36	4079763.41	113326.7614
4	Loan	36	3827286.23	106313.5064
5	Finance	31	3051189.27	98425.46032
6	IT	25	2940035.78	117601.4312

Report Explanation:

- **What it shows:** For each department — number of active employees, total monthly salary bill, and average salary per employee.
- **Why it matters:** Salary is the largest operational expense for most banks. Understanding where the HR budget is spent helps identify overstaffed or underpaid departments.
- **How it helps management or operations:** If Customer Service has the highest employee count but low average salary, they may be junior staff needing training. If IT has a high average salary but few employees, they may be understaffed.

Link technical findings to business decisions:

- `sum(e.salary)` shows total cost. If Operations department has high total salary but low deposits per employee, consider automation.
- `avg(e.salary)` per department. If Sales department has lowest average salary but highest revenue generation, they may be undercompensated.

Business Value & Recommendation:

The output shows which departments consume most HR budget. If Customer Service has highest employee count but Report 8 shows many unresolved fraud alerts needing cross-department coordination, recommend: shift budget from hiring more junior staff to hiring fewer but experienced staff. If IT has high average salary but system outages occur frequently, recommend: conduct skills audit or consider outsourcing specific functions.

Report 14: Cross-Selling Opportunity - Customers Without Credit Cards

Report Name: Cross-Selling Opportunity - Customers Without Credit Cards

Report Summary: This query shows top 10 active customers with high balance, providing a short, actionable lead list for the sales team to offer credit cards.

Query:

```
select top 10
c.customer_id,
c.first_name,
c.last_name,
sum(a.balance) as total_balance
from Customers c
join Accounts_backup a on c.customer_id =
a.customer_id
join Account_Status ast on a.status = ast.status_id
where ast.status_name = 'Active'
group by c.customer_id, c.first_name, c.last_name
having sum(a.balance) > 50000
order by total_balance desc;
```

Query Output:

	customer_id	first_name	last_name	total_balance
1	4694	Irya	Edwin	27769391.57
2	640	Leela	Sura	17229173.82
3	728	Joshua	Bhatia	15576186.67
4	4702	Charita	Reddy	15535928.91
5	2573	Tamanna	Suresh	13589600.53
6	53	Ishanvi	Wadhwa	13428671.76
7	1861	Indrajit	Bhattacharyy a	12254643.31
8	1149	Zehaan	Seth	11625855.67
9	23	Liam	Choudhary	11598869.59
10	1516	Wahab	Vora	11565255.97

Report Explanation:

- **What it shows:** Top 10 active customers with total balance above 50,000 BDT. Credit card ownership is not checked here to keep query simple and output short.
- **Why it matters:** Marketing and sales teams need a short, actionable list. Too many rows become overwhelming and unused.
- **How it helps management or operations:** Sales team gets 10 high-value customers to contact. After finishing, run again for the next 10.

Link technical findings to business decisions:

- TOP 10 with ORDER BY total_balance DESC puts highest-value customers first.
- HAVING sum(a.balance) > 50000 filters to meaningful prospects only.
- Credit card status can be checked manually by the sales team when contacting customers.

Business Value & Recommendation:

The output shows the top 10 high-balance active customers. Recommend: sales team contacts these 10 customers to offer credit cards. Track how many accept. If conversion rate is high (e.g., 7 out of 10), increase TOP to 20 next month. If conversion is low (e.g., 2 out of 10), add a promotional offer (waive first year fee, cashback on first purchase) before contacting the next batch. Run this report monthly to refresh the lead list.

7. Conclusion

The BRAC Bank database audit and normalization project successfully transformed a flawed, inconsistent dataset into a clean, normalized, and analysis-ready relational structure. By identifying and resolving structural issues, data quality violations, and normalization anomalies across 22+ tables, we established referential integrity, created lookup tables to achieve 2NF, and enforced primary and foreign key relationships throughout the schema. The 14 analytical reports generated from the corrected database provide actionable insights into operational performance, financial trends, customer behavior, risk detection, and cross-selling opportunities, directly supporting management decision-making. This project demonstrates that systematic data cleaning and normalization are essential prerequisites for reliable business intelligence, and the resulting database now serves as a solid foundation for BRAC Bank's ongoing reporting, fraud monitoring, and customer relationship management initiatives.

8. ERD

